

Name: Satyanarayanan Seshadri

Email: satya@iitm.ac.in

Shell Chair Professor – since 2024

Associate Professor, Department of Applied Mechanics and Biomedical Engg. IITM

Academic Qualification

Degree/award	Year	Discipline/field	Organisation and country
Ph.D.	2007	Mech. Eng	Texas A&M University, College Station, TX, US
M.S.	2004	Mech. Eng	Texas A&M University, College Station, TX, US
B.E.	2002	Mech. Eng	University of Madras, Chennai, India

Experience

Position held	Organisation	Department/Centre	Start	End
Associate Prof.	IIT Madras	Applied Mechanics	2021	Current
Head	IIT Madras	Energy Consortium	2022	Current
Assoc. Head	IIT Madras	School of Sustainability	2024	Current
Assistant Prof.	IIT Madras	Applied Mechanics	2015	2021
R&D Head	Forbes Marshall	Chennai Exploratory R&D	2013	2015
Lead Engineer	General Electric	Global Research Center	2008	2013
Postdoc. Assoc	Rutgers Univ.	Environmental Science	2007	2008

Current Centres (where Satya Seshadri is the PI/Head/faculty-in-charge)

- Kotak IIT Madras Save Energy Mission (KISEM): 2022 to Current
 - Established 7 assessment centres at IITM, IITB, IITRpr, IITGn, IITH, IITG & IITI
 - 200+ Energy assessments and audits in MSME across 30 sectors
 - Over INR 180 Cr in saving recommendations and 60+ crores in realised savings
- The Energy Consortium: 2021 to Current
 - India's first industry-academia-government consortium with 12 paid industry members
 - 17 core faculty members, 36 patents, 279 publications and 33 sponsored and consultancy projects, 9 CSR projects and 5 start-ups
- Shell IITM Centre for Energy Research (SICER): 2024 to Current
 - Over 15 projects, with focus on decarbonisation
 - Four pillars of engagement – Research, Consortium, Chair professor and Start-ups
- Centre of Excellence in Energy and Telecom (CEET), an IITM Sec. 8 company
 - Focus on translation research in energy storage, motors and controllers, IoT and communication and EV, with a staff of over 60 members and full-time CEO

- b. Established facilities for battery recycling, thermal energy storage, 1 MWh battery storage for the grid, solar PV recycling

Key Projects:

1. Development of Steam Generation Heat pumps (UAY Phase, 2019 – 2022)
 - a. Developed and deployed India's first steam generation heat pump
 - b. Commercialised through TrigenDC, an IITM Incubated Start-up
2. Development of Steam Generation Expander (DST SERI and SERD 2018 – 2024)
 - a. Developed and launched patented steam expander for pressure recovery from process steam
 - b. Commercialised through Wankel Energy Systems, an IITM Incubated Start-up
3. Pilot Demonstration of residential heat pumps in Ladakh (2023 – 2024)
 - a. Designed an ultra-low ambient heat pump for deployment in Leh/Ladakh region
 - b. Deployed single dwelling (2.5 kW) and multi dwelling (28 kW) units to operate down to temperature levels of – 35° C.
4. Design of CCHP system based on solid oxide fuel cells (2023 – 2025)
 - a. Evaluating optimum design of SOFC system design to achieve greater than 95% system efficiency.
 - b. Performing design integration with data centres for reducing its carbon footprint.

Awards:

- 1.) Shell Chair Professor – 2024, Corpus fund for Shell IITM Centre for Energy Research
- 2.) CSR Changemaker Award – 2023 for excellence in Kotak IIT Madras Save Energy Mission (KISEM)
- 3.) Young leaders' delegation to Israel 2018, Innovation Fellow nominated from IITM to join 12 other leaders from across the country

Service to the Institute:

- 1.) Advisor, Global Research Partnership at IITM Advisor, Global Research Partnership (Office of Global Engagement) at IITM since 2023
- 2.) Faculty-in-charge, Deakin University-IITM Research Academy Since 2024
- 3.) Admissions Committee, MS (Entrepreneurship) 2021-2024
1. Deputy advisor, Centre for Innovation (CFI) and faculty-in-charge Nirmaan 2017-2022
2. Advisor, Corporate Relations (Office of Alumni and Corporate Relations) 2021-2023
3. Board Member, Industrial Consultancy and Sponsored Research, IITM since 2024

Start-ups:

1. **2022:** Wankel Energy Systems Pvt. Ltd, co-founder: Focused on improving steam systems utilisation by recovering wasted potential across PRV (**MIT Clean Energy Prize 2024 finalist**)
2. **2023:** TrigenDC, co-founder: Electrification of Industrial Heating by developing high temperature and steam generation heat pumps. (**FLCTD award winner in 2022 for technology of 120 C Heat pump**)

3. **2024:** EnergyEta. Co-founder: Building Energy Efficiency operating system using IoT, Physics Informed ML for monitoring, control and optimisation industrial and commercial energy systems. Awarded RBDSAI IITM start-up grant for Building Energy Efficiency
4. **2024:** AranTree Consulting, Co-founder: ESG, Sustainability and decarbonisation technology consulting
- 4.) Deputy advisor, Centre for Innovation (CFI)

List of Peer reviewed Q1 Journal articles

1. Koundinya S, Seshadri S, Hafner A. Numerical investigation of high temperature heat pump integrated hybrid cooling system. *Journal of Building Engineering*. 2024 Oct 11; 98:111037. IF6.7
2. Selvaraj R, Seshadri S, Nagendra SS, Vasa NJ. Hollow-Core Fiber Based Broadband Absorption Spectroscopy with Dual-Wavelength Measurements to Remove the Effect of Multi-Gas Interference. *IEEE Sensors Letters*. 2024 Jul 12; 8(8):3426971. IF2.2
3. Sanyal S, Bhuyan P, Karthikeyan R, Alroy R, Siva Kumar G, Mandal S, Kamaraj M, Seshadri S, Sarma VS. High Temperature Air and Steam Oxidation and Fireside Corrosion Behavior of 304HCu Stainless Steel: Dichotomous Role of Grain Boundary Engineering. *High Temperature Corrosion of Materials*. 2024 Jun 28:1-26. IF2.2
4. Banerjee N, Tropea C, Seshadri S. Methodology to compute spray cooling in the nucleate boiling regime. *International Journal of Heat and Mass Transfer*. 2024 Jun 15; 225:125375. IF5
5. Banerjee N, Tropea C, Seshadri S. Methodology for modeling spray cooling of a cylindrical tube heated in the film boiling regime. *International Journal of Multiphase Flow*. 2024 Jan 1; 171:104662. IF3.6
6. Koundinya S, Jothilingam J, Seshadri S. Low-pressure steam generating heat pump–A design and field implementation case study. *Thermal Science and Engineering Progress*. 2023 Oct 1; 45:102140. IF5.1
7. Sankaralingam RK, Seshadri S, Sunarso J. Parametric optimisation using data model to improve the energy efficiency of vanadium redox flow batteries. *Journal of Energy Storage*. 2023 Aug 1; 64:107163. IF8.9
8. Chandra M, Seshadri S, Vasa NJ. Dual-wavelength absorption technique for dryness measurement of wet steam. *Applied Optics*. 2023 Apr 10; 62(11):2748-55. IF1.7
9. Kumar KS, Vidhya YE, Selvaraj R, Seshadri S, Nagendra SS, Vasa NJ. Dual absorption broadband photoacoustic technique to eliminate interference in gas mixtures. *IEEE Sensors Journal*. 2023 Feb 2; 23(6):5703-12. IF4.3
10. Karthikeyan R, Seshadri S, Sarma VS, Kamaraj M. Long-exposure air and steam oxidation characteristics of in 617 alloys. *Corrosion and Materials Degradation*. 2023 Jan 31;4(1):90-103. IF3.4
11. Mukherjee A, Seshadri S. Numerical Study on the effect of port geometry of intake manifold in a Steam Wankel Expander. *Thermal Science and Engineering Progress*. 2023 Jan 1; 37:101621. IF5.1

12. Koundinya S, Seshadri S. Energy, exergy, environmental, and economic (4E) analysis and selection of best refrigerant using TOPSIS method for industrial heat pumps. *Thermal Science and Engineering Progress*. 2022 Dec 1; 36:101491. IF5.1
13. Rajavelu H, Vasa NJ, Seshadri S. Hollow-core optical fiber-based laser-induced breakdown spectroscopy technique for the elemental analysis of pulverized coal. *Applied Physics A*. 2022 Oct; 128(10):868. IF2.5
14. Gopal VV, Seshadri S. Effect of cut-off and compression ratio on the isentropic efficiency during off-design and part-load operations of a Wankel rotary steam expander used for small scale cogeneration. *Applied Thermal Engineering*. 2022 May 5; 207:118212. IF6.1
15. Sankaralingam RK, Seshadri S, Sunarso J, Bhatt AI, Kapoor A. Effect of electrolyte parameters on the discharge characteristics of planar zinc-air flow battery with polymer gel electrolyte as separator. *Energy Storage*. 2022 Apr; 4(2):e304. IF3.6
16. Surendran A, Seshadri S. A novel transcritical-recuperative two-stage Organic Rankine Cycle for dual/multi-source heat recovery applications. *Energy*. 2022 Mar 1; 242:122961. IF9.0
17. Sankaralingam RK, Seshadri S, Sunarso J, Bhatt AI, Kapoor A. PVA-based KOH polymer gel electrolyte as a membrane separator for zinc-air flow battery. *Materials Today: Proceedings*. 2022 Jan 1; 64:1649-54. IF4.9
18. Surendran A, Seshadri S. An ejector based Transcritical Regenerative Series Two-Stage Organic Rankine Cycle for dual/multi-source heat recovery applications. *Thermal Science and Engineering Progress*. 2022 Jan 1; 27:101158. IF5.1
19. Sankaralingam RK, Seshadri S, Sunarso J, Bhatt AI, Kapoor A. Overview of the factors affecting the performance of vanadium redox flow batteries. *Journal of Energy Storage*. 2021 Sep 1; 41:102857. IF8.9
20. Rajavelu H, Vasa NJ, Seshadri S. Laser-induced breakdown spectroscopy combined with temporal plasma analysis of C2 molecular emission for carbon analysis in coal. *Applied Spectroscopy*. 2021 Jul; 75(7):893-900. IF2.2
21. Rajavelu H, Vasa NJ, Seshadri S. LIBS technique combined with blow gas and vacuum suction to remove particle cloud and enhance emission intensity during characterization of powder samples. *Spectrochimica Acta Part B: Atomic Spectroscopy*. 2021 Jul 1; 181:106215. IF3.2
22. Surendran A, Seshadri S. Design and performance analysis of a novel Transcritical Regenerative Series Two stage Organic Rankine Cycle for dual source waste heat recovery. *Energy*. 2020 Jul 15; 203:117800. IF9.0
23. Dogra V, Seshadri S. A polarization-resolved light scattering method for eliminating the interference of water aerosol in industrial stack PM measurement. *Aerosol Science and Technology*. 2020 Jun 2; (6):729-38. IF2.8

24. Rajavelu H, Vasa NJ, Seshadri S. Effect of ambiance on the coal characterization using laser-induced breakdown spectroscopy (LIBS). *Applied Physics A*. 2020 Jun; 126:1-0. IF2.5
25. Surendran A, Seshadri S. Performance investigation of two stage Organic Rankine Cycle (ORC) architectures using induction turbine layouts in dual source waste heat recovery. *Energy Conversion and Management: X*. 2020 Apr 1; 6:100029. IF7.1
26. Mallik AK, Sarma TP, Roy A, Panchagnula MV, Seshadri S. Phase Doppler particle analyser (PDPA) characterization and modeling of sprays from orthogonally interacting water and air jets. *Journal of Flow Visualization and Image Processing*. 2020; 27(2) IF0.8
27. Mainelis G, Seshadri S, Garbuzenko OB, Han T, Wang Z, Minko T. Characterization and application of a nose-only exposure chamber for inhalation delivery of liposomal drugs and nucleic acids to mice. *Journal of aerosol medicine and pulmonary drug delivery*. 2013 Dec 1; 26(6):345-54. PMID: 23530772. IF2.1
28. Seshadri S, Han T, Krumins V, Fennell DE, Mainelis G. Application of ATP bioluminescence method to characterize performance of bioaerosol sampling devices. *Journal of Aerosol Science*. 2009 Feb 1; 40(2):113-21. IF3.9
29. Seshadri S, Hari S, Hu S, McFarland AR. A circumferential slot in-line virtual impactor. *Aerosol science and technology*. 2008 Jan 1; 42(1):40-9. IF2.8
30. Hu S, Seshadri S, McFarland AR. CFD study on compound impaction in a jet-in-well impactor. *Aerosol science and technology*. 2007 Nov 9; 41(12):1102-9. IF2.8
31. McFarland A, McIntyre P, Seshadri S. Virtual impact fractionation of superconductor precursor powders in inert gas atmosphere. *IEEE transactions on applied superconductivity*. 2007 Jun; 17(2):2576-9. IF1.7
32. Seshadri S, Phares DJ, Kim T, Kihm KD, Smith DD, McIntyre PM. Performance of a slit virtual impactor operated at high particle mass loading. *Journal of aerosol science*. 2005 Apr 1; 36(4):541-7. IF3.9
33. Kim TK, Seshadri S, Kihm KD, Phares DJ, McIntyre PM. Visualization of defect particle transmission to the major flow of a slit virtual impactor. *Aerosol science and technology*. 2004 Sep 1; 38(9):870-80. IF2.8

LIST OF CONFERENCES

- 1: Koushik AV, Sankaralingam RK, Paudel S, Neupane N, Saranathan S, Raghunath D, Magesh NN, Kulkarni S, Sivaprasad S, Navagana GS, Seshadri S, Aghalayam P, Rengaswamy R. Climate Action Tool (CAT): Philosophy and case studies. In 2024 AIChE Annual Meeting 2024 October 27. AIChE.
- 2: Banerjee N, Tropea C, Seshadri S. Methodology to compute spray cooling in the nucleate boiling regime. In 16th Triennial International Conference on Liquid Atomization and Spray Systems, Jun 23-27, 2024, Shanghai, China.

- 3: Banerjee N, Tropea C, Seshadri S. Methodology for modelling spray cooling of a cylindrical tube heated in the film boiling regime. In Proceedings of the 9th World Congress on Momentum, Heat and Mass Transfer, April – 2024, London, United Kingdom.
- 4: Neupane N, Bashyal B, Koundinya S, Saranathan S, Seshadri S. ML Based Predictive Modeling of Air-side Heat Transfer Coefficient in Finned-Tube Heat Exchangers. In International Conference on Fluid, Thermal, and Energy Systems, June 6-8, 2024, NIT Calicut.
- 5: Dhipankumar Kanakasabapathi, Sumant Bansal, Dhanavardhan M, Prof. Satyanarayan Seshadri, Prof. Santosh Kumar Sahu, “Estimation of LCOE and LCCA for a Renewable Energy and Energy Efficiency Project in India’s Paper Industry: A comparison between Rooftop Solar and Steam Expander for Power Generation “, IEEE Power & Energy Society Innovative Smart Grid Technologies Europe 2024.
- 6: Nivash M, Koundinya S, Sankaralingam RK, Neupane N, Seshadri S. Machine Learning based Predictive Modeling of Heat Transfer Coefficient in Two-phase Flows of Refrigerants. In International Conference on Refrigeration and Air-conditioning, 13-15 March 2024, IIT Madras Chennai.
- 7: Dhanavardhan M, Neupane N, Koundinya S, Sankaralingam RK, Martin SR, Seshadri S. An Experimental Study using THF Hydrates as a Thermal Energy Storage Material for Milk-Chilling Application. In International Conference on Refrigeration and Air-conditioning, 13-15 March 2024, IIT Madras Chennai.
- 8: Shiva N, Banerjee N, Seshadri S. Comparative study on Multiphase Dynamics of Unsteady Film Condensation over a Vertical Plate in different regimes. In Proceedings of the 27th National and 5th International ISHMT-ASTFE Heat and Mass Transfer Conference December 14-17, 2023, IIT Patna, Patna-801106, Bihar, India 2024. Begel House Inc.
- 9: Koundinya S, Jothilingam J, Martin SR, Seshadri S. Numerical and Experimental Study of 120° C Heat Pump Using a Scroll Compressor. In International Conference on Compressors and their Systems 2023 Sep 11 (pp. 327-338). Cham: Springer Nature Switzerland.
- 10: Kafayat AK, Prof. Satyanarayan Seshadri, Prof. Santosh Kumar Sahu. The Interrelationship Between Carbon dioxide Emission, Population, Energy Consumption and Economic Growth in Ghana: A Review. In International Conference on Energy Conversion and Storage, 18-20 January 2023, IIT Madras Chennai, India.
- 11: Singh HP, Prof. Satyanarayan Seshadri on International Conference on Energy Conversion and Storage (IECS-2023): Numerical Study to Determine the Optimum Flow Field Design for Tubular Vanadium Redox Flow Battery, 18-20 January 2023, IIT Madras Chennai, India.
- 12: Mukherjee, Auronil and Seshadri, Satyanarayanan, "Numerical Investigation of Pressure Losses and its Effect During Intake in a Steam Wankel Expander" (2022). International Compressor Engineering Conference. Paper 2747. <https://docs.lib.purdue.edu/icec/2747>
- 13: Shiva N, Banerjee N, Seshadri S. Multiphase Modelling of Thin Film Flow Over the Vertical Plate Under Gravity in Pseudo-Laminar Region. In Conference on Fluid Mechanics and Fluid Power 2022 Dec 14 (pp. 131-146). Singapore: Springer Nature Singapore.

- 14: Banerjee N, Seshadri S. A Mathematical Approach for prediction of Spray Heat Transfer for Various Textured Cylindrical Surfaces in Nucleate Boiling regime. In Conference on ILASS-Asia 22nd Annual Conference on Liquid Atomization and Spray Systems, Oct 28-30, 2022, IIT Indore.
- 15: Chandra M, Seshadri S, Vasa NJ. Michelson Interferometry technique for the Dryness measurement of Saturated Steam. In Propagation Through and Characterization of Atmospheric and Oceanic Phenomena 2022 Jul 11 (pp. JW2A-13). Optica Publishing Group.
- 16: Karthikeyan R, Seshadri S, Sarma VS, Kamaraj M. Long exposure steam oxidation studies on 304HCu and IN 617 for ultra-supercritical condition, 2nd Corrosion and Materials Degradation Web Conference, 2022 Jul 5. CMDWC.
- 17: Koundinya S, S. Seshadri, Performance study of desiccant enhanced ejector-based transcritical CO₂ heat pump technology for high-efficiency space cooling., in: 15th IIR-Gustav Lorentzen Conf. Nat. Refrig. (GL2022), 13-15th June 2022, NTNU, Trondheim, Norway. Proc., 2022. <http://dx.doi.org/10.18462/iir.gl2022.0156>.
- 18: Mukherjee, Auronil and Seshadri, Satyanarayanan, "Numerical Investigation of Pressure Losses and its Effect During Intake in a Steam Wankel Expander" (2022). International Compressor Engineering Conference. Paper 2747. <https://docs.lib.purdue.edu/icec/2747>
- 19: Singh P, Srikanti K, Chandra M, Gopalan R, Seshadri S. Experimental Study to Enhance the Performance of Li-Fe-Co-Si Active Magnetic Regenerator for Room Temperature Cooling Applications.
- 20: R. K. Sankaralingam, S. Seshadri, J. Sunarso, A.I. Bhatt, A. Kapoor, PVA-based KOH polymer gel electrolyte as a membrane separator for zinc-air flow battery, 1st International conference on energy materials, 5th to 8th November, 2021. Materials Today: Proceedings Volume 64, Part 4, 2022, Pages 1649-1654. <https://doi.org/10.1016/j.matpr.2022.05.222>
- 21: Emmanuel A, Koundinya. S, Seshadri. S. Development of Heat Exchanger Models for Predicting Heat Transfer Behaviour of Mixed Refrigerants, virtual seminar in Applied Mechanics, 30-31st July 2021, Lecture Notes in Mechanical Engineering, (2022), 687-696. https://doi.org/10.1007/978-981-16-9539-1_51
- 22: Koundinya. S, Seshadri. S. A novel desiccant-based cooling system for hot and humid climates, virtual seminar in Applied Mechanics, 30-31st July 2021, Lecture Notes in Mechanical Engineering, (2022), 673-685. https://doi.org/10.1007/978-981-16-9539-1_50
- 23: Devi RG, Seshadri S, Michelassi V. Numerical Model for Condensing Steam Through Labyrinth Seal. In Turbo Expo: Power for Land, Sea, and Air 2021 Jun 7 (Vol. 85017, p. V008T22A007). American Society of Mechanical Engineers.
- 24: Rajavelu H, Vasa NJ, Seshadri S. Determination of Ash Content in Coal Using Laser-Induced Breakdown Spectroscopy with Multivariate Analysis. In ICOL-2019: Proceedings of the International Conference on Optics and Electro-Optics, Dehradun, India 2021 (pp. 399-402). Springer Singapore.
- 25: Bhat A, Surendran A, Seshadri S. Preliminary Investigations of a Novel Dual Evaporator Dual Ejector Refrigeration Cycle, 2021.

- 26: R. K. Sankaralingam, S. Seshadri. Effect of electrolyte flow rate and different anode materials on the discharge performance of flow type zinc-air. International Flow Battery Forum, 2020 (19th-21st Jan 2021).
- 27: Chandra M, Seshadri S, Vasa NJ. Real-time Steam Quality Measurement using Tunable Diode Laser Absorption Spectroscopy for Process Industries. In Applied Industrial Spectroscopy 2020 Jun 22 (pp. AM3F-2). Optica Publishing Group.
- 28: Balachandran A, Seshadri S. Application of continuous wavelet transform for elimination of spectral interference in UV spectroscopy. In Applied Industrial Spectroscopy 2020 Jun 22 (pp. ATu4I-6). Optica Publishing Group.
- 29: Singh S, Hafner A, Banasiak K, Seshadri S, Maiya P, Smitt SM, Gabriell CH. Heat pump/chiller system for centralized kitchens in India. In Proceedings of the 14th IIR-Gustav Lorentzen Conference on Natural Refrigerants 2020. IIR.
- 28: Banerjee N, Seshadri S. Dynamics of Thin Film over Flat Rough Surface Moving with a Specified Velocity. In Conference on Fluid Mechanics and Fluid Power Dec 9-11, 2020, IIT Guwahati, Guwahati-781039, Assam, India.
- 30: Surendran A, Seshadri S, Seshadri S. Performance analysis and optimization of R245fa-cyclopentane zeotropic mixtures in partial evaporating organic Rankine cycles (PEORC) for low temperature heat recovery. In 5th International Seminar on ORC Power Systems 2019.
- 31: Hemalaxmi R, Aparna N, Vasa NJ, Seshadri S. Nanosecond and femtosecond laser induced breakdown spectroscopic studies of coal and ash. In 2018 Conference on Lasers and Electro-Optics Pacific Rim (CLEO-PR) 2018 Jul 29 (pp. 1-2). IEEE.
- 32: Chatterjee A, Bernal E, Seshadri S, Mayer O, Greaves M, Dimroth F, Kurtz S, Sala G, Bett AW. Linear Fresnel reflector based solar radiation concentrator for combined heating and power. In AIP Conference Proceedings-American Institute of Physics 2011 Dec 1 (Vol. 1407, No. 1, p. 257).

Book Chapters:

1. Ram Kishore Sankaralingam, Manjari Ravuthan, Satyanarayanan Seshadri, Jaka Sunarso, Elsevier, June 2024, Nanostructured Materials Engineering and Characterization for Battery Applications, Chapter 2 - Battery modeling, Pages 43-74, ISBN 978-0323913041, <https://doi.org/10.1016/B978-0-323-91304-1.00002-2>.
2. Subburaj G, Gopal VV, Seshadri S., Springer Verlag, Singapore, April 2022, In Recent Advances in Applied Mechanics: Proceedings of Virtual Seminar on Applied Mechanics (VSAM 2021), Comparative Study of 2D Heat Transfer Models for a Wankel Expander, Pages 663-671. ISBN 978-9811695414
3. Emmanuel A, Seshadri S, Koundinya S., Springer Verlag, Singapore, April 2022, In Recent Advances in Applied Mechanics: Proceedings of Virtual Seminar on Applied Mechanics (VSAM 2021), Development of Heat Exchanger Models for Predicting Heat Transfer Behaviour of Mixed Refrigerants, Pages 687-696. ISBN 978-9811695414

4. Koundinya S, Seshadri S., Springer Verlag, Singapore, April 2022, In Recent Advances in Applied Mechanics: Proceedings of Virtual Seminar on Applied Mechanics (VSAM 2021), A Novel Desiccant-Based Cooling System for Hot and Humid Climates, 673-685. ISBN 978-9811695414

Membership of Prestigious Academies and Committees Note:

1. Member of Technical Committee under “Offset Mechanism of Carbon Credit Trading Scheme, Bureau of Energy Efficiency”, August 2024 to Present
2. Expert Member of Committee “Recovery and Recycling of End-of-Life Solar PV Panels/Modules”, Department of Science and Technology, Nov 2024 to Present
3. Vaishwik Bharatiya Vaigyanik (Vaibhav) Summit under “Energy Vertical and Advanced Fossil and Nuclear Technologies Horizontal – Host”, October 2020